

The Notebooks · Era III — Relate

Graph paper / coordinate lounge

Testing Mode · UK · IB · US

Exercises

T.III.1.001 · Standard form A×10ⁿ introduction (KS3) · core · UK_KS3,IB_MYP,US_CCSS_68

- (a) [never/neutral/1mk] Write 5700000 in standard form.
- (b) [never/UK/1mk] Write 6.5×10^5 as an ordinary number.
- (c) [never/UK/3mk] Calculate $(5.7 \times 10^6) \times (3.0 \times 10^2)$ in standard form.

T.III.1.002 · Convert ordinary ↔ standard form fluency · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Write 36000 in standard form.
- (b) [never/UK/1mk] Write 4.0×10^2 as an ordinary number.
- (c) [never/UK/3mk] Calculate $(3.6 \times 10^4) \times (2.2 \times 10^2)$ in standard form.

T.III.1.003 · Order numbers in standard form · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Write 2500000 in standard form.
- (b) [never/UK/1mk] Write 7.5×10^2 as an ordinary number.
- (c) [never/UK/3mk] Calculate $(2.5 \times 10^6) \times (1.8 \times 10^2)$ in standard form.

T.III.1.004 · Calculate with standard form (non-calc) · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Write 0.0064 in standard form.
- (b) [never/UK/1mk] Write 3.1×10^3 as an ordinary number.
- (c) [never/UK/3mk] Calculate $(6.4 \times 10^2) \times (2.4 \times 10^2)$ in standard form.

T.III.1.005 · Standard form with calculator (GCSE) · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Write 9.2×10^{-5} in standard form.
- (b) [never/UK/1mk] Write 3.3×10^3 as an ordinary number.
- (c) [gdc/UK/3mk ■ LAB] GDC lab for «Standard form with calculator (GCSE)»: record menu path and numerical output.

T.III.1.006 · Negative numbers in financial contexts · core · UK_KS3,IB_MYP,US_CCSS_68

- (a) [never/neutral/1mk] Write 24000 in standard form.
- (b) [never/UK/1mk] Write 3.2×10^4 as an ordinary number.
- (c) [never/UK/3mk] Calculate $(2.4 \times 10^4) \times (1.3 \times 10^2)$ in standard form.

T.III.1.007 · Recurring decimal notation (dot notation UK) · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Write 3000 in standard form.
- (b) [never/UK/1mk] Write 4.4×10^4 as an ordinary number.
- (c) [never/UK/3mk] Calculate $(3.0 \times 10^3) \times (2.6 \times 10^2)$ in standard form.

T.III.1.008 · Convert recurring decimals to fractions · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Write 7.4×10^{-5} in standard form.
- (b) [never/UK/1mk] Write 2.9×10^4 as an ordinary number.
- (c) [never/UK/3mk] Calculate $(7.4 \times 10^2) \times (2.1 \times 10^2)$ in standard form.

T.III.1.009 · Absolute value equations simple (KS3) · core · UK_KS3,IB_MYP,US_CCSS_68

- (a) [never/neutral/1mk] Write 6.7×10^{-5} in standard form.
- (b) [never/UK/1mk] Write 4.1×10^3 as an ordinary number.
- (c) [never/UK/3mk] Calculate $(6.7 \times 10^2) \times (2.6 \times 10^2)$ in standard form.

T.III.1.010 · Error intervals from rounded values intro · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/2mk] 190 cm to nearest 10 cm. Write the error interval for length L.
- (b) [never/UK/2mk] $w=2.6$ cm, $l=11.6$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/UK/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.III.1.011 · Upper and lower bounds of rounded measurements · core · UK_GCSE_H

- (a) [never/neutral/2mk] 180 cm to nearest 10 cm. Write the error interval for length L.
- (b) [never/UK/2mk] $w=5.5$ cm, $l=5.1$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/UK/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.III.1.012 · Bounds in calculations (sum/product/quotient) · challenge · UK_GCSE_H

- (a) [never/neutral/2mk] 70 cm to nearest 10 cm. Write the error interval for length L.
- (b) [never/UK/2mk] $w=4.6$ cm, $l=7.3$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/UK/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.III.1.013 · Limits of accuracy wording exam pack · core · UK_GCSE_H

- (a) [never/neutral/2mk] 40 cm to nearest 10 cm. Write the error interval for length L.
- (b) [never/UK/2mk] $w=4.5$ cm, $l=9.8$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/UK/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.III.1.014 · Bounds to 1 d.p. from truncated measurements · challenge · UK_GCSE_H

- (a) [never/neutral/2mk] 90 cm to nearest 10 cm. Write the error interval for length L.
- (b) [never/UK/2mk] $w=6.3$ cm, $l=10.8$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/UK/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.III.1.015 · Error intervals for continuous measures · core · UK_GCSE_H

- (a) [never/neutral/2mk] 150 cm to nearest 10 cm. Write the error interval for length L.
- (b) [never/UK/2mk] $w=6.1$ cm, $l=8.0$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/UK/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.III.1.016 · Bounds: upper and lower for truncated data · challenge · UK_GCSE_H

- (a) [never/neutral/2mk] 60 cm to nearest 10 cm. Write the error interval for length L.
- (b) [never/UK/2mk] $w=5.9$ cm, $l=6.3$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/UK/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.III.1.017 · Calculator vs non-calculator standard form · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Write $3.8e-05$ in standard form.
- (b) [never/UK/1mk] Write 6.0×10^2 as an ordinary number.
- (c) [gdc/UK/3mk ■ LAB] GDC lab for «Calculator vs non-calculator standard form»: record menu path and numerical output.

T.III.1.018 · Standard form applied science context 1 · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Write 14000 in standard form.
- (b) [never/UK/1mk] Write 8.1×10^4 as an ordinary number.
- (c) [never/UK/3mk] Calculate $(1.4 \times 10^4) \times (2.2 \times 10^2)$ in standard form.

T.III.1.019 · Standard form applied science context 2 · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Write 260000 in standard form.
- (b) [never/UK/1mk] Write 7.1×10^2 as an ordinary number.
- (c) [never/UK/3mk] Calculate $(2.6 \times 10^5) \times (2.7 \times 10^2)$ in standard form.

T.III.1.020 · Standard form applied science context 3 · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Write 5500 in standard form.
- (b) [never/UK/1mk] Write 5.8×10^5 as an ordinary number.
- (c) [never/UK/3mk] Calculate $(5.5 \times 10^3) \times (1.1 \times 10^2)$ in standard form.

T.III.2.001 · Compound measure: speed (m/s and km/h) · core · UK_KS3,IB_MYP,US_CCSS_68

- (a) [allowed/neutral/2mk] Share £52 in the ratio 5:8. First share?
- (b) [allowed/UK/1mk] Ratio 5:8; first quantity 20. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 8 cm. Real distance in km?

T.III.2.002 · Compound measure: density (mass÷volume) · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/2mk] Share £130 in the ratio 5:8. First share?
- (b) [allowed/UK/1mk] Ratio 5:8; first quantity 50. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 8 cm. Real distance in km?

T.III.2.003 · Compound measure: pressure · core · UK_GCSE_H

- (a) [allowed/neutral/2mk] Share £64 in the ratio 3:5. First share?
- (b) [allowed/UK/1mk] Ratio 3:5; first quantity 24. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 6 cm. Real distance in km?

T.III.2.004 · Rates of pay and unit pricing · core · UK_KS3,IB_MYP,US_CCSS_68

- (a) [allowed/neutral/2mk] Share £56 in the ratio 3:5. First share?
- (b) [allowed/UK/1mk] Ratio 3:5; first quantity 21. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 6 cm. Real distance in km?

T.III.2.005 · Currency conversion with exchange rates · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/2mk] Share £65 in the ratio 5:8. First share?
- (b) [allowed/UK/1mk] Ratio 5:8; first quantity 25. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 8 cm. Real distance in km?

T.III.2.006 · Map scales as ratios (1:n) · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/2mk] Share £20 in the ratio 2:3. First share?
- (b) [allowed/UK/1mk] Ratio 2:3; first quantity 8. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 5 cm. Real distance in km?

T.III.2.007 · Sharing in a ratio three-part · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/2mk] Share £130 in the ratio 5:8. First share?
- (b) [allowed/UK/1mk] Ratio 5:8; first quantity 50. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 8 cm. Real distance in km?

T.III.2.008 · Ratio given difference or total variants · challenge · UK_GCSE_H

- (a) [allowed/neutral/2mk] Share £88 in the ratio 3:5. First share?
- (b) [allowed/UK/1mk] Ratio 3:5; first quantity 33. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 6 cm. Real distance in km?

T.III.2.009 · Direct proportion graphs through origin · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/2mk] Share £20 in the ratio 2:3. First share?
- (b) [allowed/UK/1mk] Ratio 2:3; first quantity 8. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 5 cm. Real distance in km?

T.III.2.010 · Inverse proportion intro ($xy=k$) · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/2mk] Share £112 in the ratio 3:5. First share?
- (b) [allowed/UK/1mk] Ratio 3:5; first quantity 42. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 6 cm. Real distance in km?

T.III.2.011 · Speed–distance–time with unit changes · challenge · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/2mk] Share £143 in the ratio 4:7. First share?
- (b) [allowed/UK/1mk] Ratio 4:7; first quantity 52. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 7 cm. Real distance in km?

T.III.2.012 · Population density and area ratios · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/2mk] Share £50 in the ratio 2:3. First share?
- (b) [allowed/UK/1mk] Ratio 2:3; first quantity 20. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 5 cm. Real distance in km?

T.III.2.013 · Recipe scaling and mixture ratios · core · UK_KS3,IB_MYP,US_CCSS_68

- (a) [allowed/neutral/2mk] Share £78 in the ratio 5:8. First share?
- (b) [allowed/UK/1mk] Ratio 5:8; first quantity 30. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 8 cm. Real distance in km?

T.III.2.014 · CCSS RP: tape diagrams for percent/ratio · core · US_CCSS_68,UK_KS3,IB_MYP

- (a) [allowed/neutral/2mk] Share £55 in the ratio 4:7. First share?
- (b) [allowed/US/1mk] Ratio 4:7; first quantity 20. Second?
- (c) [allowed/US/2mk] Map scale 1:25000. Map distance 7 cm. Real distance in km?

T.III.2.015 · CCSS RP: percent as ratio of part to whole · core · US_CCSS_68,UK_KS3,IB_MYP

- (a) [allowed/neutral/2mk] Share £32 in the ratio 3:5. First share?
- (b) [allowed/US/1mk] Ratio 3:5; first quantity 12. Second?
- (c) [allowed/US/2mk] Map scale 1:25000. Map distance 6 cm. Real distance in km?

T.III.2.016 · Mixing problems with ratios · challenge · UK_GCSE_H

- (a) [allowed/neutral/2mk] Share £35 in the ratio 2:3. First share?
- (b) [allowed/UK/1mk] Ratio 2:3; first quantity 14. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 5 cm. Real distance in km?

T.III.2.017 · Scale drawings measure and interpret · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/2mk] Share £121 in the ratio 4:7. First share?
- (b) [allowed/UK/1mk] Ratio 4:7; first quantity 44. Second?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Scale drawings measure and interpret»: build the figure, measure, verify the claimed relation.

T.III.2.018 · Proportion word problems multi-step · challenge · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/2mk] Share £182 in the ratio 5:8. First share?
- (b) [allowed/UK/1mk] Ratio 5:8; first quantity 70. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 8 cm. Real distance in km?

T.III.2.019 · Average speed for multi-leg journeys · challenge · UK_GCSE_H

- (a) [allowed/neutral/2mk] Share £78 in the ratio 5:8. First share?
- (b) [allowed/UK/1mk] Ratio 5:8; first quantity 30. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 8 cm. Real distance in km?

T.III.2.020 · Convert compound units step by step · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/2mk] Share £121 in the ratio 4:7. First share?
- (b) [allowed/UK/1mk] Ratio 4:7; first quantity 44. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 7 cm. Real distance in km?

T.III.2.021 · Compound measures mixed drill 1 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/2mk] Share £112 in the ratio 3:5. First share?
- (b) [allowed/UK/1mk] Ratio 3:5; first quantity 42. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 6 cm. Real distance in km?

T.III.2.022 · Compound measures mixed drill 2 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/2mk] Share £72 in the ratio 3:5. First share?
- (b) [allowed/UK/1mk] Ratio 3:5; first quantity 27. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 6 cm. Real distance in km?

T.III.2.023 · Compound measures mixed drill 3 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/2mk] Share £112 in the ratio 3:5. First share?
- (b) [allowed/UK/1mk] Ratio 3:5; first quantity 42. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 6 cm. Real distance in km?

T.III.2.024 · Compound measures mixed drill 4 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/2mk] Share £32 in the ratio 3:5. First share?
- (b) [allowed/UK/1mk] Ratio 3:5; first quantity 12. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 6 cm. Real distance in km?

T.III.2.025 · Compound measures mixed drill 5 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/2mk] Share £169 in the ratio 5:8. First share?
- (b) [allowed/UK/1mk] Ratio 5:8; first quantity 65. Second?
- (c) [allowed/UK/2mk] Map scale 1:25000. Map distance 8 cm. Real distance in km?

T.III.3.001 · Find original value after % change (reverse %) · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Increase £400 by 8%.
- (b) [allowed/UK/2mk] After a 8% increase a price is £432. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 8% then by another 8%. Overall % increase?

T.III.3.002 · Simple interest exam pack ($I = \frac{PRT}{100}$) · core · UK_KS3,IB_MYP,US_CCSS_68

- (a) [allowed/neutral/1mk] Increase £400 by 12%.
- (b) [allowed/UK/2mk] After a 12% increase a price is £448. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 12% then by another 12%. Overall % increase?

T.III.3.003 · Compound interest (non-calc appreciation) · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Increase £400 by 8%.
- (b) [never/UK/2mk] After a 8% increase a price is £432. Find the original.
- (c) [never/UK/3mk] A quantity rises by 8% then by another 8%. Overall % increase?

T.III.3.004 · Compound interest depreciation / decay · core · UK_GCSE_H

- (a) [allowed/neutral/1mk] Increase £120 by 10%.
- (b) [allowed/UK/2mk] After a 10% increase a price is £132. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 10% then by another 10%. Overall % increase?

T.III.3.005 · VAT and sales tax multi-step · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Increase £250 by 10%.
- (b) [allowed/UK/2mk] After a 10% increase a price is £275. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 10% then by another 10%. Overall % increase?

T.III.3.006 · Hire purchase and interest comparisons · challenge · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Increase £80 by 15%.
- (b) [allowed/UK/2mk] After a 15% increase a price is £92. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 15% then by another 15%. Overall % increase?

T.III.3.007 · Successive percentage changes (not additive) · challenge · UK_GCSE_H

- (a) [allowed/neutral/1mk] Increase £80 by 5%.
- (b) [allowed/UK/2mk] After a 5% increase a price is £84. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 5% then by another 5%. Overall % increase?

T.III.3.008 · Express one quantity as % of another · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Increase £200 by 5%.
- (b) [allowed/UK/2mk] After a 5% increase a price is £210. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 5% then by another 5%. Overall % increase?

T.III.3.009 · Percentage profit and loss · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Increase £200 by 12%.
- (b) [allowed/UK/2mk] After a 12% increase a price is £224. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 12% then by another 12%. Overall % increase?

T.III.3.010 · Repeated percentage growth GDC workflow · core · IB_AI_SL,IB_AI_HL,IB_AI_HL

- (a) [gdc/neutral/1mk] Increase £400 by 8%.
- (b) [gdc/IB/2mk] After a 8% increase a price is £432. Find the original.
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «Repeated percentage growth GDC workflow»: record menu path and numerical output.

T.III.3.011 · Bank statements and balance tracking · core · UK_KS3,IB_MYP,US_CCSS_68

- (a) [allowed/neutral/1mk] Increase £120 by 12%.
- (b) [allowed/UK/2mk] After a 12% increase a price is £134. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 12% then by another 12%. Overall % increase?

T.III.3.012 · Best-buy percentage vs absolute saving · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Increase £80 by 12%.
- (b) [allowed/UK/2mk] After a 12% increase a price is £89. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 12% then by another 12%. Overall % increase?

T.III.3.013 · Percentage multipliers one-step fluency · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Increase £250 by 20%.
- (b) [allowed/UK/2mk] After a 20% increase a price is £300. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 20% then by another 20%. Overall % increase?

T.III.3.014 · Find % when given two related quantities · challenge · UK_GCSE_H

- (a) [allowed/neutral/1mk] Increase £400 by 20%.
- (b) [allowed/UK/2mk] After a 20% increase a price is £480. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 20% then by another 20%. Overall % increase?

T.III.3.015 · Inflation and real-terms value (simple) · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] Increase £80 by 10%.
- (b) [allowed/UK/2mk] After a 10% increase a price is £88. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 10% then by another 10%. Overall % increase?

T.III.3.016 · Express changes using percentage multipliers chain · challenge · UK_GCSE_H

- (a) [allowed/neutral/1mk] Increase £80 by 8%.
- (b) [allowed/UK/2mk] After a 8% increase a price is £86. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 8% then by another 8%. Overall % increase?

T.III.3.017 · Financial % multi-step scenario 1 · challenge · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Increase £250 by 5%.
- (b) [allowed/UK/2mk] After a 5% increase a price is £262. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 5% then by another 5%. Overall % increase?

T.III.3.018 · Financial % multi-step scenario 2 · challenge · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Increase £80 by 5%.
- (b) [allowed/UK/2mk] After a 5% increase a price is £84. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 5% then by another 5%. Overall % increase?

T.III.3.019 · Financial % multi-step scenario 3 · challenge · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Increase £250 by 5%.
- (b) [allowed/UK/2mk] After a 5% increase a price is £262. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 5% then by another 5%. Overall % increase?

T.III.3.020 · Financial % multi-step scenario 4 · challenge · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Increase £250 by 8%.
- (b) [allowed/UK/2mk] After a 8% increase a price is £270. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 8% then by another 8%. Overall % increase?

T.III.3.021 · Financial % multi-step scenario 5 · challenge · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Increase £80 by 15%.
- (b) [allowed/UK/2mk] After a 15% increase a price is £92. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 15% then by another 15%. Overall % increase?

T.III.4.001 · Laws of indices exam pack (integer) · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Simplify $\sqrt{50}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{2}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{2}+2)(\sqrt{2}-2)$.

T.III.4.002 · Fractional indices as roots · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Simplify $\sqrt{20}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{3}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{3}+2)(\sqrt{3}-2)$.

T.III.4.003 · Negative and fractional indices combined · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Simplify $\sqrt{72}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{2}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{2}+2)(\sqrt{2}-2)$.

T.III.4.004 · Surds: simplify $\sqrt{(ab)}$ and $\sqrt{(a/b)}$ · core · UK_GCSE_H

- (a) [never/neutral/1mk] Simplify $\sqrt{50}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{2}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{2}+2)(\sqrt{2}-2)$.

T.III.4.005 · Surds: expand and simplify $(a+\sqrt{b})(c+\sqrt{d})$ · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Simplify $\sqrt{18}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{3}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{3}+2)(\sqrt{3}-2)$.

T.III.4.006 · Rationalise denominators (monomial and binomial) · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Simplify $\sqrt{27}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{7}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{7}+2)(\sqrt{7}-2)$.

T.III.4.007 · Scientific notation vs UK standard form wording · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Write 0.0079 in standard form.
- (b) [never/UK/1mk] Write 3.4×10^5 as an ordinary number.
- (c) [never/UK/3mk] Calculate $(7.9 \times 10^2) \times (1.6 \times 10^2)$ in standard form.

T.III.4.008 · Estimate powers and roots without calc · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Simplify $\sqrt{32}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{3}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{3}+2)(\sqrt{3}-2)$.

T.III.4.009 · Zero and negative indices meaning pack · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Simplify $\sqrt{27}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{7}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{7}+2)(\sqrt{7}-2)$.

T.III.4.010 · Work in standard form then convert back · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Write 270000 in standard form.
- (b) [never/UK/1mk] Write 3.0×10^4 as an ordinary number.
- (c) [never/UK/3mk] Calculate $(2.7 \times 10^5) \times (2.5 \times 10^2)$ in standard form.

T.III.4.011 · Evaluate with fractional/negative indices non-calc · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Simplify $\sqrt{8}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{7}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{7}+2)(\sqrt{7}-2)$.

T.III.5.001 · Expand single and double brackets exam pack · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Expand $2(x+4)$.
- (b) [never/UK/2mk] Expand $(x+5)(x+5)$.
- (c) [never/UK/2mk] Factorise $x^2 + 10x + 25$.

T.III.5.002 · Factorise linear expressions fully · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Expand $4(x+5)$.
- (b) [never/UK/2mk] Expand $(x+3)(x+3)$.
- (c) [never/UK/2mk] Factorise $x^2 + 6x + 9$.

T.III.5.003 · Factorise quadratics a=1 exam pack · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Expand $4(x+4)$.
- (b) [never/UK/2mk] Expand $(x+4)(x+3)$.
- (c) [never/UK/2mk] Factorise $x^2 + 7x + 12$.

T.III.5.004 · Factorise difference of two squares · core · UK_GCSE_H

- (a) [never/neutral/1mk] Simplify $\sqrt{8}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{3}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{3}+2)(\sqrt{3}-2)$.

T.III.5.005 · Algebraic fractions simplify (linear) · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Expand $2(x+3)$.
- (b) [never/UK/2mk] Expand $(x+2)(x+4)$.
- (c) [never/UK/2mk] Factorise $x^2 + 6x + 8$.

T.III.5.006 · Write expressions from worded contexts · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Expand $3(x+2)$.
- (b) [never/UK/2mk] Expand $(x+2)(x+3)$.
- (c) [never/UK/2mk] Factorise $x^2 + 5x + 6$.

T.III.5.007 · Identities vs equations recognition · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Expand $3(x+2)$.
- (b) [never/UK/2mk] Expand $(x+1)(x+3)$.
- (c) [never/UK/2mk] Factorise $x^2 + 4x + 3$.

T.III.5.008 · Collect like terms with multiple variables · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Expand $5(x+1)$.
- (b) [never/UK/2mk] Expand $(x+3)(x+6)$.
- (c) [never/UK/2mk] Factorise $x^2 + 9x + 18$.

T.III.5.009 · Substitution including negatives and powers · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Expand $5(x+1)$.
- (b) [never/UK/2mk] Expand $(x+1)(x+2)$.
- (c) [never/UK/2mk] Factorise $x^2 + 3x + 2$.

T.III.5.010 · Expand three binomials (Higher) · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Expand $5(x+4)$.
- (b) [never/UK/2mk] Expand $(x+1)(x+2)$.
- (c) [never/UK/2mk] Factorise $x^2 + 3x + 2$.

T.III.5.011 · Factorise by grouping (four terms) · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Expand $6(x+1)$.
- (b) [never/UK/2mk] Expand $(x+4)(x+2)$.
- (c) [never/UK/2mk] Factorise $x^2 + 6x + 8$.

T.III.6.001 · Solve linear equations with fractions · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Solve $6x + 8 = 38$.
- (b) [never/UK/2mk] Solve $6(x - 8) = -12$.
- (c) [never/UK/2mk] List integers n satisfying $5 < n \leq 9$.

T.III.6.002 · Form and solve equations from geometry · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Solve $4x + 7 = 31$.
- (b) [never/UK/2mk] Solve $4(x - 7) = 0$.
- (c) [never/UK/2mk] List integers n satisfying $6 < n \leq 10$.

T.III.6.003 · Inequalities on number lines (UK open/closed) · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Solve $9x + 5 = 32$.
- (b) [never/UK/2mk] Solve $9(x - 5) = -9$.
- (c) [never/UK/2mk] List integers n satisfying $3 < n \leq 7$.

T.III.6.004 · Solve linear inequalities and list integers · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Solve $7x + 4 = 67$.
- (b) [never/UK/2mk] Solve $7(x - 4) = 42$.
- (c) [never/UK/2mk] List integers n satisfying $9 < n \leq 13$.

T.III.6.005 · Simultaneous equations elimination (KS3/4) · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Solve $2x + 3 = 13$.
- (b) [never/UK/2mk] Solve $2(x - 3) = 6$.
- (c) [never/UK/2mk] List integers n satisfying $5 < n \leq 9$.

T.III.6.006 · Simultaneous equations substitution · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Solve $4x + 2 = 22$.
- (b) [never/UK/2mk] Solve $4(x - 2) = 16$.
- (c) [never/UK/2mk] List integers n satisfying $5 < n \leq 9$.

T.III.6.007 · Graphical solution of simultaneous equations · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Solve $5x + 1 = 46$.
- (b) [never/UK/2mk] Solve $5(x - 1) = 45$.
- (c) [allowed/UK/3mk ■ LAB] Desmos lab for «Graphical solution of simultaneous equations»: vary one parameter, capture before/after, write one observation.

T.III.6.008 · Equations with unknown on both sides · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Solve $8x + 2 = 42$.
- (b) [never/UK/2mk] Solve $8(x - 2) = 32$.
- (c) [never/UK/2mk] List integers n satisfying $5 < n \leq 9$.

T.III.6.009 · Inequalities with negative coefficients · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Solve $2x + 6 = 24$.
- (b) [never/UK/2mk] Solve $2(x - 6) = 8$.
- (c) [never/UK/2mk] List integers n satisfying $9 < n \leq 13$.

T.III.6.010 · Quadratic inequalities intro via graphs · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Solve $4x + 6 = 10$.
- (b) [never/UK/2mk] Solve $4(x - 6) = -16$.
- (c) [allowed/UK/3mk ■ LAB] Desmos lab for «Quadratic inequalities intro via graphs»: vary one parameter, capture before/after, write one observation.

T.III.6.011 · Trial and improvement (legacy dialect) · strange · UK_GCSE_H

- (a) [never/neutral/1mk] Solve $3x + 6 = 18$.
- (b) [never/UK/2mk] Solve $3(x - 6) = -3$.
- (c) [never/UK/2mk] List integers n satisfying $4 < n \leq 8$.

T.III.7.001 · Gradient from coordinates (UK wording) · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Expand $2(x+5)$.
- (b) [never/UK/2mk] Expand $(x+3)(x+4)$.
- (c) [never/UK/2mk] Factorise $x^2 + 7x + 12$.

T.III.7.002 · Equation of a line $y=mx+c$ fluency · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Expand $2(x+1)$.
- (b) [never/UK/2mk] Expand $(x+5)(x+6)$.
- (c) [never/UK/2mk] Factorise $x^2 + 11x + 30$.

T.III.7.003 · Parallel and perpendicular gradients · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Expand $6(x+6)$.
- (b) [never/UK/2mk] Expand $(x+2)(x+1)$.
- (c) [never/UK/2mk] Factorise $x^2 + 3x + 2$.

T.III.7.004 · Midpoint and length of a line segment · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Expand $5(x+2)$.
- (b) [never/UK/2mk] Expand $(x+4)(x+5)$.
- (c) [never/UK/2mk] Factorise $x^2 + 9x + 20$.

T.III.7.005 · Real graphs: conversion and distance–time · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Expand $4(x+1)$.
- (b) [never/UK/2mk] Expand $(x+2)(x+3)$.
- (c) [allowed/UK/3mk ■ LAB] Graph lab for «Real graphs: conversion and distance–time»: plot, mark key points, state the trend.

T.III.7.006 · Speed from distance–time graph gradients · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Expand $5(x+6)$.
- (b) [never/UK/2mk] Expand $(x+5)(x+5)$.
- (c) [allowed/UK/3mk ■ LAB] Graph lab for «Speed from distance–time graph gradients»: plot, mark key points, state the trend.

T.III.7.007 · Instantaneous rate of change on a curve (KS4 tease) · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Expand $4(x+5)$.
- (b) [never/UK/2mk] Expand $(x+1)(x+4)$.
- (c) [allowed/UK/3mk ■ LAB] Desmos lab for «Instantaneous rate of change on a curve (KS4 tease)»: vary one parameter, capture before/after, write one observation.

T.III.7.008 · Find equation of parallel line through a point · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Expand $4(x+5)$.
- (b) [never/UK/2mk] Expand $(x+3)(x+6)$.
- (c) [never/UK/2mk] Factorise $x^2 + 9x + 18$.

T.III.7.009 · Interpret intercepts in context · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Expand $2(x+5)$.
- (b) [never/UK/2mk] Expand $(x+4)(x+1)$.
- (c) [allowed/UK/3mk ■ LAB] Graph lab for «Interpret intercepts in context»: plot, mark key points, state the trend.

T.III.7.010 · Solve linear equations graphically on Desmos · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Expand $5(x+3)$.
- (b) [never/UK/2mk] Expand $(x+5)(x+4)$.
- (c) [allowed/UK/3mk ■ LAB] Desmos lab for «Solve linear equations graphically on Desmos»: vary one parameter, capture before/after, write one observation.

T.III.7.011 · Find intercepts algebraically and graphically · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Expand $5(x+5)$.
- (b) [never/UK/2mk] Expand $(x+1)(x+2)$.
- (c) [allowed/UK/3mk ■ LAB] Desmos lab for «Find intercepts algebraically and graphically»: vary one parameter, capture before/after, write one observation.

T.III.8.001 · Plans and elevations of 3D shapes · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] [Plans and elevations of 3D shapes] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $7 \times 3 \times 5$: how many rectangles in a correct net?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Plans and elevations of 3D shapes»: build the figure, measure, verify the claimed relation.

T.III.8.002 · Nets of pyramids, prisms, cylinders · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] A cuboid measures $8 \times 7 \times 6$. How many faces does a correct net contain?
- (b) [never/UK/2mk] Which of these cannot fold to a cube: cross of 6 squares vs row of 6 squares? Explain.
- (c) [never/UK/2mk] Surface area of cuboid $8 \times 7 \times 6$?

T.III.8.003 · Bearings intro: three-figure bearings · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Write 255° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 255 . Bearing of A from B?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Bearings intro: three-figure bearings»: build the figure, measure, verify the claimed relation.

T.III.8.004 · Construct triangles given SSS/SAS/ASA · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] [Construct triangles given SSS/SAS/ASA] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $3 \times 3 \times 3$: how many rectangles in a correct net?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Construct triangles given SSS/SAS/ASA»: build the figure, measure, verify the claimed relation.

T.III.8.005 · Loci: set of points equidistant · core · UK_GCSE_H

- (a) [never/neutral/1mk] [Loci: set of points equidistant] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $4 \times 7 \times 5$: how many rectangles in a correct net?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Loci: set of points equidistant»: build the figure, measure, verify the claimed relation.

T.III.8.006 · Loci: regions satisfying inequalities · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] [Loci: regions satisfying inequalities] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $5 \times 3 \times 5$: how many rectangles in a correct net?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Loci: regions satisfying inequalities»: build the figure, measure, verify the claimed relation.

T.III.8.007 · Pythagoras exam worded (2D) · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Legs 6 cm, 8 cm. Hypotenuse?
- (b) [never/UK/2mk] Hypotenuse 10 m, leg 6 m. Other leg?
- (c) [allowed/UK/2mk] Ladder 10 m; base 6 m from wall. Reach height?

T.III.8.008 · Pythagoras in 3D (space diagonal) · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Legs 6 cm, 8 cm. Hypotenuse?
- (b) [never/UK/2mk] Hypotenuse 10 m, leg 6 m. Other leg?
- (c) [allowed/UK/2mk] Ladder 10 m; base 6 m from wall. Reach height?

T.III.8.009 · Circle area/circumference exact answers in π · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] [Circle area/circumference exact answers in π] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $9 \times 4 \times 2$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 9) standing on its base.

T.III.8.010 · Arc length and sector area (GCSE) · core · UK_GCSE_H

- (a) [never/neutral/1mk] [Arc length and sector area (GCSE)] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $3 \times 7 \times 4$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 3) standing on its base.

T.III.8.011 · Volume/surface area formula selection pack · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] [Volume/surface area formula selection pack] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $9 \times 6 \times 6$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 9) standing on its base.

T.III.8.012 · Similar shapes length/area/volume scale factors · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Lengths scale by 3. Area scale factor?
- (b) [never/UK/1mk] A side 6 cm becomes ? after enlargement $SF=3$.
- (c) [never/UK/2mk] Volumes scale by ? if lengths scale by 3.

T.III.8.013 · Congruence conditions SSS SAS ASA RHS · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Name three congruence conditions for triangles.
- (b) [never/UK/2mk] Does SSA always prove congruence? Why/why not?
- (c) [never/UK/2mk] In «Congruence conditions SSS SAS ASA RHS»: what does CPCTC allow you to conclude after triangles are congruent?

T.III.8.014 · Enlargement with fractional and negative SF · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Lengths scale by 5. Area scale factor?
- (b) [never/UK/1mk] A side 6 cm becomes ? after enlargement $SF=5$.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Enlargement with fractional and negative SF»: build the figure, measure, verify the claimed relation.

T.III.8.015 · Describe transformations fully (GCSE language) · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] [Describe transformations fully (GCSE language)] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $8 \times 4 \times 6$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 8) standing on its base.

T.III.8.016 · Interior/exterior angles of polygons exam · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] [Interior/exterior angles of polygons exam] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $8 \times 6 \times 6$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 8) standing on its base.

T.III.8.017 · Tessellation angle conditions · strange · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] [Tessellation angle conditions] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $7 \times 2 \times 4$: how many rectangles in a correct net?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Tessellation angle conditions»: build the figure, measure, verify the claimed relation.

T.III.8.018 · Surface area of cylinders exact and decimal · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] [Surface area of cylinders exact and decimal] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $6 \times 5 \times 2$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 6) standing on its base.

T.III.8.019 · Exact trig values preview: 30–60–90 links · core · UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.III.8.020 · Isometric drawing of 3D shapes · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] [Isometric drawing of 3D shapes] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $8 \times 2 \times 4$: how many rectangles in a correct net?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Isometric drawing of 3D shapes»: build the figure, measure, verify the claimed relation.

T.III.8.021 · Exact answers leave in terms of π pack · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] [Exact answers leave in terms of π pack] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $9 \times 4 \times 2$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 9) standing on its base.

T.III.8.022 · Surface area of spheres leave in π · core · UK_GCSE_H

- (a) [never/neutral/1mk] [Surface area of spheres leave in π] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $6 \times 5 \times 5$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 6) standing on its base.

T.III.8.023 · Voronoi diagrams nearest-site partitioning · strange · IB_AI_HL

- (a) [never/neutral/1mk] [Voronoi diagrams nearest-site partitioning] Which view is the plan — front, side, or top?
- (b) [never/IB/1mk] Cuboid $5 \times 2 \times 4$: how many rectangles in a correct net?
- (c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Voronoi diagrams nearest-site partitioning»: build the figure, measure, verify the claimed relation.

T.III.8.024 · Voronoi applications: services catchment · strange · IB_AI_HL

- (a) [never/neutral/1mk] [Voronoi applications: services catchment] Which view is the plan — front, side, or top?
- (b) [never/IB/1mk] Cuboid $4 \times 2 \times 3$: how many rectangles in a correct net?
- (c) [allowed/IB/2mk] Describe the front elevation of a square-based pyramid (base 4) standing on its base.

T.III.8.025 · Graph theory: walks trails paths cycles (light) · strange · IB_AI_HL

- (a) [never/neutral/1mk] [Graph theory: walks trails paths cycles (light)] Which view is the plan — front, side, or top?
- (b) [never/IB/1mk] Cuboid $3 \times 7 \times 3$: how many rectangles in a correct net?
- (c) [allowed/IB/2mk] Describe the front elevation of a square-based pyramid (base 3) standing on its base.

T.III.8.026 · Shortest path / Chinese Postman light · strange · IB_AI_HL

- (a) [never/neutral/1mk] [Shortest path / Chinese Postman light] Which view is the plan — front, side, or top?
- (b) [never/IB/1mk] Cuboid $8 \times 4 \times 6$: how many rectangles in a correct net?
- (c) [allowed/IB/2mk] Describe the front elevation of a square-based pyramid (base 8) standing on its base.

T.III.8.027 · Plans elevations practice figure 1 · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] [Plans elevations practice figure 1] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $3 \times 2 \times 6$: how many rectangles in a correct net?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Plans elevations practice figure 1»: build the figure, measure, verify the claimed relation.

T.III.8.028 · Plans elevations practice figure 2 · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] [Plans elevations practice figure 2] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $9 \times 4 \times 3$: how many rectangles in a correct net?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Plans elevations practice figure 2»: build the figure, measure, verify the claimed relation.

T.III.8.029 · Plans elevations practice figure 3 · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] [Plans elevations practice figure 3] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $3 \times 6 \times 5$: how many rectangles in a correct net?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Plans elevations practice figure 3»: build the figure, measure, verify the claimed relation.

T.III.9.001 · Venn diagrams for probability (two sets) · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Bag: 8 red, 4 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.III.9.002 · Venn diagrams three sets · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Bag: 3 red, 2 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.III.9.003 · Frequency trees for probability · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Mean of 4, 10, 11, 11, 15, 18?
- (b) [allowed/UK/1mk] Range of 4, 10, 11, 11, 15, 18?
- (c) [allowed/UK/2mk] Mean of five numbers is 11. Four are 4, 10, 11, 11. Find the fifth.

T.III.9.004 · Histograms with unequal class widths · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Class width 5, frequency 15. Frequency density?
- (b) [never/UK/1mk] A histogram bar has width 5 and FD 3.0. Frequency?
- (c) [never/UK/2mk] Why must unequal-width histograms use frequency density, not raw frequency for bar height?

T.III.9.005 · Cumulative frequency curves and medians · core · UK_GCSE_H

- (a) [allowed/neutral/1mk] Mean of 4, 5, 5, 11, 12, 13?
- (b) [allowed/UK/1mk] Range of 4, 5, 5, 11, 12, 13?
- (c) [allowed/UK/3mk ■ LAB] Graph lab for «Cumulative frequency curves and medians»: plot, mark key points, state the trend.

T.III.9.006 · Box plots from CF and compare distributions · core · UK_GCSE_H

- (a) [allowed/neutral/1mk] Mean of 3, 4, 5, 9, 9, 14?
- (b) [allowed/UK/1mk] Range of 3, 4, 5, 9, 9, 14?
- (c) [allowed/UK/2mk] Mean of five numbers is 12. Four are 3, 4, 5, 9. Find the fifth.

T.III.9.007 · Product rule for counting (KS4 Higher) · core · UK_GCSE_H

- (a) [allowed/neutral/1mk] Mean of 5, 7, 10, 15, 15, 17?
- (b) [allowed/UK/1mk] Range of 5, 7, 10, 15, 15, 17?
- (c) [allowed/UK/2mk] Mean of five numbers is 12. Four are 5, 7, 10, 15. Find the fifth.

T.III.9.008 · Sample space diagrams for two events · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Mean of 4, 7, 9, 10, 12, 13?
- (b) [allowed/UK/1mk] Range of 4, 7, 9, 10, 12, 13?
- (c) [allowed/UK/2mk] Mean of five numbers is 9. Four are 4, 7, 9, 10. Find the fifth.

T.III.9.009 · Relative frequency vs theoretical probability · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Mean of 5, 6, 6, 8, 15, 15?
- (b) [allowed/UK/1mk] Range of 5, 6, 6, 8, 15, 15?
- (c) [allowed/UK/2mk] Mean of five numbers is 8. Four are 5, 6, 6, 8. Find the fifth.

T.III.9.010 · Capture–recapture estimation (UK contexts) · strange · UK_GCSE_H

- (a) [allowed/neutral/1mk] Mean of 4, 6, 12, 12, 12, 12?
- (b) [allowed/UK/1mk] Range of 4, 6, 12, 12, 12, 12?
- (c) [allowed/UK/2mk] Mean of five numbers is 11. Four are 4, 6, 12, 12. Find the fifth.

T.III.9.011 · Stratified sampling calculations · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] Mean of 4, 4, 12, 17, 18, 18?
- (b) [allowed/UK/1mk] Range of 4, 4, 12, 17, 18, 18?
- (c) [allowed/UK/2mk] Mean of five numbers is 14. Four are 4, 4, 12, 17. Find the fifth.

T.III.9.012 · Two-way tables probability conditional · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] Mean of 3, 4, 5, 8, 11, 14?
- (b) [allowed/UK/1mk] Range of 3, 4, 5, 8, 11, 14?
- (c) [allowed/UK/2mk] Mean of five numbers is 14. Four are 3, 4, 5, 8. Find the fifth.

T.III.9.013 · Pie chart construction from data · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Mean of 5, 10, 11, 14, 15, 16?
- (b) [allowed/UK/1mk] Range of 5, 10, 11, 14, 15, 16?
- (c) [allowed/UK/2mk] Mean of five numbers is 11. Four are 5, 10, 11, 14. Find the fifth.

T.III.9.014 · Moving averages (time series intro) · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] Mean of 5, 5, 9, 12, 15, 16?
- (b) [allowed/UK/1mk] Range of 5, 5, 9, 12, 15, 16?
- (c) [allowed/UK/2mk] Mean of five numbers is 10. Four are 5, 5, 9, 12. Find the fifth.

T.III.9.015 · CCSS SP: chance device simulations middle school · core · US_CCSS_68,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Mean of 3, 3, 6, 7, 13, 14?
- (b) [allowed/US/1mk] Range of 3, 3, 6, 7, 13, 14?
- (c) [allowed/US/2mk] Mean of five numbers is 10. Four are 3, 3, 6, 7. Find the fifth.

T.III.9.016 · CCSS SP: bivariate categorical associations · core · US_CCSS_68,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Mean of 3, 5, 14, 16, 17, 18?
- (b) [allowed/US/1mk] Range of 3, 5, 14, 16, 17, 18?
- (c) [allowed/US/2mk] Mean of five numbers is 13. Four are 3, 5, 14, 16. Find the fifth.

T.III.9.017 · Mean from frequency table (estimated) · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Mean of 3, 8, 9, 12, 15, 16?
- (b) [allowed/UK/1mk] Range of 3, 8, 9, 12, 15, 16?
- (c) [allowed/UK/2mk] Mean of five numbers is 8. Four are 3, 8, 9, 12. Find the fifth.

T.III.9.018 · Modal class and median class from grouped data · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Mean of 3, 6, 6, 13, 14, 17?
- (b) [allowed/UK/1mk] Range of 3, 6, 6, 13, 14, 17?
- (c) [allowed/UK/2mk] Mean of five numbers is 9. Four are 3, 6, 6, 13. Find the fifth.

T.III.9.019 · Probability scale 0 to 1 language · core · UK_KS3,IB_MYP,US_CCSS_68

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Probability scale 0 to 1 language»?
- (b) [allowed/UK/2mk] Write a justification sentence frame for «Probability scale 0 to 1 language»: Because ____, by ____, therefore ____.
- (c) [allowed/UK/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Probability scale 0 to 1 language», then interpret.

T.III.9.020 · Expectation from theoretical probability · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Mean of 4, 5, 6, 8, 10, 11?
- (b) [allowed/UK/1mk] Range of 4, 5, 6, 8, 10, 11?
- (c) [allowed/UK/2mk] Mean of five numbers is 13. Four are 4, 5, 6, 8. Find the fifth.

T.III.9.021 · AP Stats: describing distributions SOCS · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] Mean of 9, 12, 13, 14, 17, 18?
- (b) [allowed/US/1mk] Range of 9, 12, 13, 14, 17, 18?
- (c) [allowed/US/2mk] Mean of five numbers is 14. Four are 9, 12, 13, 14. Find the fifth.

T.III.9.022 · AP Stats: comparing dual boxplots FRQ · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «AP Stats: comparing dual boxplots FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «AP Stats: comparing dual boxplots FRQ»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP Stats: comparing dual boxplots FRQ», then interpret.

T.III.9.023 · IB AI: boxplots and outliers 1.5×IQR rule · core · IB_AI_SL,IB_AI_HL,AP_STATS

- (a) [allowed/neutral/1mk] Mean of 3, 4, 12, 16, 17, 18?
- (b) [allowed/IB/1mk] Range of 3, 4, 12, 16, 17, 18?
- (c) [allowed/IB/2mk] Mean of five numbers is 9. Four are 3, 4, 12, 16. Find the fifth.

T.III.9.024 · Misleading scales and truncated axes critique · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Mean of 3, 5, 6, 9, 12, 17?
- (b) [allowed/UK/1mk] Range of 3, 5, 6, 9, 12, 17?
- (c) [allowed/UK/2mk] Mean of five numbers is 10. Four are 3, 5, 6, 9. Find the fifth.

T.III.9.025 · Cumulative frequency ↔ box plot pipeline · core · UK_GCSE_H

- (a) [allowed/neutral/1mk] Mean of 3, 6, 7, 11, 11, 12?
- (b) [allowed/UK/1mk] Range of 3, 6, 7, 11, 11, 12?
- (c) [allowed/UK/3mk ■ LAB] Graph lab for «Cumulative frequency ↔ box plot pipeline»: plot, mark key points, state the trend.

T.III.9.026 · Experimental probability long-run estimate · core · UK_KS3,IB_MYP,US_CCSS_68

- (a) [never/neutral/1mk] Mean of 3, 4, 6, 7, 10, 15?
- (b) [never/UK/1mk] Range of 3, 4, 6, 7, 10, 15?
- (c) [never/UK/2mk] Mean of five numbers is 14. Four are 3, 4, 6, 7. Find the fifth.

T.III.9.027 · Questionnaire bias and leading questions · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] Mean of 3, 10, 11, 13, 17, 17?
- (b) [allowed/US/1mk] Range of 3, 10, 11, 13, 17, 17?
- (c) [allowed/US/2mk] Mean of five numbers is 8. Four are 3, 10, 11, 13. Find the fifth.

T.III.9.028 · Frequency polygon from histogram · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Class width 10, frequency 24. Frequency density?
- (b) [never/UK/1mk] A histogram bar has width 10 and FD 2.4. Frequency?
- (c) [allowed/UK/3mk ■ LAB] Graph lab for «Frequency polygon from histogram»: plot, mark key points, state the trend.

T.III.9.029 · Pie chart angle calculation exam · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Mean of 6, 7, 7, 11, 13, 18?
- (b) [allowed/UK/1mk] Range of 6, 7, 7, 11, 13, 18?
- (c) [allowed/UK/2mk] Mean of five numbers is 9. Four are 6, 7, 7, 11. Find the fifth.

T.III.9.030 · Combine probabilities mutually exclusive lists · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Mean of 3, 6, 7, 14, 17, 18?
- (b) [allowed/UK/1mk] Range of 3, 6, 7, 14, 17, 18?
- (c) [allowed/UK/2mk] Mean of five numbers is 11. Four are 3, 6, 7, 14. Find the fifth.

T.III.9.031 · Sampling variability dotplot intuition · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] Mean of 7, 7, 16, 16, 17, 17?
- (b) [allowed/US/1mk] Range of 7, 7, 16, 16, 17, 17?
- (c) [allowed/US/2mk] Mean of five numbers is 12. Four are 7, 7, 16, 16. Find the fifth.

T.III.9.032 · Venn probability exam set 1 · core · UK_GCSE_H

- (a) [never/neutral/1mk] Bag: 7 red, 2 blue. P(red)?
- (b) [never/UK/2mk] Without replacement, P(both red)?
- (c) [never/UK/2mk] With replacement, P(red then blue)?

T.III.9.033 · Venn probability exam set 2 · core · UK_GCSE_H

- (a) [never/neutral/1mk] Bag: 3 red, 2 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.III.9.034 · Venn probability exam set 3 · core · UK_GCSE_H

- (a) [never/neutral/1mk] Bag: 8 red, 2 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.III.9.035 · Histogram unequal width frequency density set · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Class width 20, frequency 22. Frequency density?
- (b) [never/UK/1mk] A histogram bar has width 20 and FD 1.1. Frequency?
- (c) [never/UK/2mk] Why must unequal-width histograms use frequency density, not raw frequency for bar height?

T.III.1.021 · Bounds of a calculated perimeter/area · challenge · UK_GCSE_H

- (a) [never/neutral/2mk] 60 cm to nearest 10 cm. Write the error interval for length L.
- (b) [never/UK/2mk] $w=4.5$ cm, $l=9.6$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/UK/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.III.3.022 · Reverse percentage after successive changes · challenge · UK_GCSE_H

- (a) [allowed/neutral/1mk] Increase £200 by 15%.
- (b) [allowed/UK/2mk] After a 15% increase a price is £229. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 15% then by another 15%. Overall % increase?

Answer Key — Era III

Brief solutions. Full reasoning in machine JSON.

T.III.1.001

- (a) 5.7×10^6 — $1 \leq A < 10$.
- (b) 650000 — *Shift decimal*.
- (c) 1.71×10^9 — *Multiply A; add exponents; renormalise*.

T.III.1.002

- (a) 3.6×10^4 — $1 \leq A < 10$.
- (b) 400 — *Shift decimal*.
- (c) 7.92×10^6 — *Multiply A; add exponents; renormalise*.

T.III.1.003

- (a) 2.5×10^6 — $1 \leq A < 10$.
- (b) 750 — *Shift decimal*.
- (c) 4.5×10^8 — *Multiply A; add exponents; renormalise*.

T.III.1.004

- (a) 6.4×10^{-3} — $1 \leq A < 10$.
- (b) 3100 — *Shift decimal*.
- (c) 1.536×10^0 — *Multiply A; add exponents; renormalise*.

T.III.1.005

- (a) 9.2×10^{-5} — $1 \leq A < 10$.
- (b) 3300 — *Shift decimal*.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.III.1.006

- (a) 2.4×10^4 — $1 \leq A < 10$.
- (b) 32000 — *Shift decimal*.
- (c) 3.12×10^6 — *Multiply A; add exponents; renormalise*.

T.III.1.007

- (a) 3.0×10^3 — $1 \leq A < 10$.
- (b) 44000 — *Shift decimal*.
- (c) 7.8×10^5 — *Multiply A; add exponents; renormalise*.

T.III.1.008

- (a) 7.4×10^{-5} — $1 \leq A < 10$.
- (b) 29000 — *Shift decimal*.
- (c) 1.554×10^{-2} — *Multiply A; add exponents; renormalise*.

T.III.1.009

- (a) 6.7×10^{-5} — $1 \leq A < 10$.
- (b) 4100 — *Shift decimal*.
- (c) 1.742×10^{-2} — *Multiply A; add exponents; renormalise*.

T.III.1.010

- (a) $185 \leq L < 195$ — $\pm$ half unit.
- (b) 28.60 cm — *Use UBs in $P=2(L+W)$* .
- (c) $\text{UB}(\text{distance}) \div \text{LB}(\text{time})$ — *Max quotient: max numerator, min denominator*.

T.III.1.011

- (a) $175 \leq L < 185$ — $\pm$ half unit.
- (b) 21.40 cm — *Use UBs in $P=2(L+W)$* .
- (c) $\text{UB}(\text{distance}) \div \text{LB}(\text{time})$ — *Max quotient: max numerator, min denominator*.

T.III.1.012

- (a) $65 \leq L < 75$ — $\pm$ half unit.
- (b) 24.00 cm — *Use UBs in $P=2(L+W)$* .
- (c) $\text{UB}(\text{distance}) \div \text{LB}(\text{time})$ — *Max quotient: max numerator, min denominator*.

T.III.1.013

- (a) $35 \leq L < 45$ — $\pm$ half unit.
- (b) 28.80 cm — *Use UBs in $P=2(L+W)$* .
- (c) $\text{UB}(\text{distance}) \div \text{LB}(\text{time})$ — *Max quotient: max numerator, min denominator*.

T.III.1.014

- (a) $85 \leq L < 95$ — $\pm$ half unit.

- (b) 34.40 cm — Use UBs in $P=2(L+W)$.
 (c) $\text{UB}(\text{distance}) \div \text{LB}(\text{time})$ — Max quotient: max numerator, min denominator.

T.III.1.015

- (a) $145 \leq L < 155$ — $\pm$ half unit.
 (b) 28.40 cm — Use UBs in $P=2(L+W)$.
 (c) $\text{UB}(\text{distance}) \div \text{LB}(\text{time})$ — Max quotient: max numerator, min denominator.

T.III.1.016

- (a) $55 \leq L < 65$ — $\pm$ half unit.
 (b) 24.60 cm — Use UBs in $P=2(L+W)$.
 (c) $\text{UB}(\text{distance}) \div \text{LB}(\text{time})$ — Max quotient: max numerator, min denominator.

T.III.1.017

- (a) 3.8×10^{-5} — $1 \leq A < 10$.
 (b) 600 — Shift decimal.
 (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.III.1.018

- (a) 1.4×10^4 — $1 \leq A < 10$.
 (b) 81000 — Shift decimal.
 (c) 3.08×10^6 — Multiply A; add exponents; renormalise.

T.III.1.019

- (a) 2.6×10^5 — $1 \leq A < 10$.
 (b) 710 — Shift decimal.
 (c) 7.02×10^7 — Multiply A; add exponents; renormalise.

T.III.1.020

- (a) 5.5×10^3 — $1 \leq A < 10$.
 (b) 580000 — Shift decimal.
 (c) 6.05×10^5 — Multiply A; add exponents; renormalise.

T.III.2.001

- (a) £20 — 13 parts; each £4.
 (b) 32 — $\times 4$.
 (c) 2 km — $\text{cm} \times \text{scale} \rightarrow$ convert to km.

T.III.2.002

- (a) £50 — 13 parts; each £10.
 (b) 80 — $\times 10$.
 (c) 2 km — $\text{cm} \times \text{scale} \rightarrow$ convert to km.

T.III.2.003

- (a) £24 — 8 parts; each £8.
 (b) 40 — $\times 8$.
 (c) 1.5 km — $\text{cm} \times \text{scale} \rightarrow$ convert to km.

T.III.2.004

- (a) £21 — 8 parts; each £7.
 (b) 35 — $\times 7$.
 (c) 1.5 km — $\text{cm} \times \text{scale} \rightarrow$ convert to km.

T.III.2.005

- (a) £25 — 13 parts; each £5.
 (b) 40 — $\times 5$.
 (c) 2 km — $\text{cm} \times \text{scale} \rightarrow$ convert to km.

T.III.2.006

- (a) £8 — 5 parts; each £4.
 (b) 12 — $\times 4$.
 (c) 1.25 km — $\text{cm} \times \text{scale} \rightarrow$ convert to km.

T.III.2.007

- (a) £50 — 13 parts; each £10.
 (b) 80 — $\times 10$.
 (c) 2 km — $\text{cm} \times \text{scale} \rightarrow$ convert to km.

T.III.2.008

- (a) £33 — 8 parts; each £11.
 (b) 55 — $\times 11$.
 (c) 1.5 km — $\text{cm} \times \text{scale} \rightarrow$ convert to km.

T.III.2.009

- (a) £8 — 5 parts; each £4.
- (b) 12 — $\times 4$.
- (c) 1.25 km — $cm \times scale \rightarrow$ convert to km.

T.III.2.010

- (a) £42 — 8 parts; each £14.
- (b) 70 — $\times 14$.
- (c) 1.5 km — $cm \times scale \rightarrow$ convert to km.

T.III.2.011

- (a) £52 — 11 parts; each £13.
- (b) 91 — $\times 13$.
- (c) 1.75 km — $cm \times scale \rightarrow$ convert to km.

T.III.2.012

- (a) £20 — 5 parts; each £10.
- (b) 30 — $\times 10$.
- (c) 1.25 km — $cm \times scale \rightarrow$ convert to km.

T.III.2.013

- (a) £30 — 13 parts; each £6.
- (b) 48 — $\times 6$.
- (c) 2 km — $cm \times scale \rightarrow$ convert to km.

T.III.2.014

- (a) £20 — 11 parts; each £5.
- (b) 35 — $\times 5$.
- (c) 1.75 km — $cm \times scale \rightarrow$ convert to km.

T.III.2.015

- (a) £12 — 8 parts; each £4.
- (b) 20 — $\times 4$.
- (c) 1.5 km — $cm \times scale \rightarrow$ convert to km.

T.III.2.016

- (a) £14 — 5 parts; each £7.
- (b) 21 — $\times 7$.
- (c) 1.25 km — $cm \times scale \rightarrow$ convert to km.

T.III.2.017

- (a) £44 — 11 parts; each £11.
- (b) 77 — $\times 11$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.III.2.018

- (a) £70 — 13 parts; each £14.
- (b) 112 — $\times 14$.
- (c) 2 km — $cm \times scale \rightarrow$ convert to km.

T.III.2.019

- (a) £30 — 13 parts; each £6.
- (b) 48 — $\times 6$.
- (c) 2 km — $cm \times scale \rightarrow$ convert to km.

T.III.2.020

- (a) £44 — 11 parts; each £11.
- (b) 77 — $\times 11$.
- (c) 1.75 km — $cm \times scale \rightarrow$ convert to km.

T.III.2.021

- (a) £42 — 8 parts; each £14.
- (b) 70 — $\times 14$.
- (c) 1.5 km — $cm \times scale \rightarrow$ convert to km.

T.III.2.022

- (a) £27 — 8 parts; each £9.
- (b) 45 — $\times 9$.
- (c) 1.5 km — $cm \times scale \rightarrow$ convert to km.

T.III.2.023

- (a) £42 — 8 parts; each £14.
- (b) $70 \div 14$.
- (c) $1.5 \text{ km} \div \text{cm} \times \text{scale} \rightarrow \text{convert to km}$.

T.III.2.024

- (a) £12 — 8 parts; each £4.
- (b) $20 \div 4$.
- (c) $1.5 \text{ km} \div \text{cm} \times \text{scale} \rightarrow \text{convert to km}$.

T.III.2.025

- (a) £65 — 13 parts; each £13.
- (b) $104 \div 13$.
- (c) $2 \text{ km} \div \text{cm} \times \text{scale} \rightarrow \text{convert to km}$.

T.III.3.001

- (a) £432 — $\times 1.08$.
- (b) £400 — $\div 1.08$.
- (c) 16.64% — Multiply multipliers; convert to %.

T.III.3.002

- (a) £448 — $\times 1.12$.
- (b) £400 — $\div 1.12$.
- (c) 25.44% — Multiply multipliers; convert to %.

T.III.3.003

- (a) £432 — $\times 1.08$.
- (b) £400 — $\div 1.08$.
- (c) 16.64% — Multiply multipliers; convert to %.

T.III.3.004

- (a) £132 — $\times 1.1$.
- (b) £120 — $\div 1.1$.
- (c) 21.00% — Multiply multipliers; convert to %.

T.III.3.005

- (a) £275 — $\times 1.1$.
- (b) £250 — $\div 1.1$.
- (c) 21.00% — Multiply multipliers; convert to %.

T.III.3.006

- (a) £92 — $\times 1.15$.
- (b) £80 — $\div 1.15$.
- (c) 32.25% — Multiply multipliers; convert to %.

T.III.3.007

- (a) £84 — $\times 1.05$.
- (b) £80 — $\div 1.05$.
- (c) 10.25% — Multiply multipliers; convert to %.

T.III.3.008

- (a) £210 — $\times 1.05$.
- (b) £200 — $\div 1.05$.
- (c) 10.25% — Multiply multipliers; convert to %.

T.III.3.009

- (a) £224 — $\times 1.12$.
- (b) £200 — $\div 1.12$.
- (c) 25.44% — Multiply multipliers; convert to %.

T.III.3.010

- (a) £432 — $\times 1.08$.
- (b) £400 — $\div 1.08$.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.III.3.011

- (a) £134.40 — $\times 1.12$.
- (b) £119.64 — $\div 1.12$.
- (c) 25.44% — Multiply multipliers; convert to %.

T.III.3.012

- (a) £89.60 — $\times 1.12$.
- (b) £79.46 — $\div 1.12$.

(c) 25.44% — Multiply multipliers; convert to %.

T.III.3.013

(a) £300 — $\times 1.2$.

(b) £250 — $\div 1.2$.

(c) 44.00% — Multiply multipliers; convert to %.

T.III.3.014

(a) £480 — $\times 1.2$.

(b) £400 — $\div 1.2$.

(c) 44.00% — Multiply multipliers; convert to %.

T.III.3.015

(a) £88 — $\times 1.1$.

(b) £80 — $\div 1.1$.

(c) 21.00% — Multiply multipliers; convert to %.

T.III.3.016

(a) £86.40 — $\times 1.08$.

(b) £79.63 — $\div 1.08$.

(c) 16.64% — Multiply multipliers; convert to %.

T.III.3.017

(a) £262.50 — $\times 1.05$.

(b) £249.52 — $\div 1.05$.

(c) 10.25% — Multiply multipliers; convert to %.

T.III.3.018

(a) £84 — $\times 1.05$.

(b) £80 — $\div 1.05$.

(c) 10.25% — Multiply multipliers; convert to %.

T.III.3.019

(a) £262.50 — $\times 1.05$.

(b) £249.52 — $\div 1.05$.

(c) 10.25% — Multiply multipliers; convert to %.

T.III.3.020

(a) £270 — $\times 1.08$.

(b) £250 — $\div 1.08$.

(c) 16.64% — Multiply multipliers; convert to %.

T.III.3.021

(a) £92 — $\times 1.15$.

(b) £80 — $\div 1.15$.

(c) 32.25% — Multiply multipliers; convert to %.

T.III.4.001

(a) $5\sqrt{2}$ — Extract square factors.

(b) $\sqrt{2/2}$ — Multiply by $\sqrt{b}/\sqrt{b}$.

(c) -2 — Difference of squares.

T.III.4.002

(a) $2\sqrt{5}$ — Extract square factors.

(b) $\sqrt{3/3}$ — Multiply by $\sqrt{b}/\sqrt{b}$.

(c) -1 — Difference of squares.

T.III.4.003

(a) $6\sqrt{2}$ — Extract square factors.

(b) $\sqrt{2/2}$ — Multiply by $\sqrt{b}/\sqrt{b}$.

(c) -2 — Difference of squares.

T.III.4.004

(a) $5\sqrt{2}$ — Extract square factors.

(b) $\sqrt{2/2}$ — Multiply by $\sqrt{b}/\sqrt{b}$.

(c) -2 — Difference of squares.

T.III.4.005

(a) $3\sqrt{2}$ — Extract square factors.

(b) $\sqrt{3/3}$ — Multiply by $\sqrt{b}/\sqrt{b}$.

(c) -1 — Difference of squares.

T.III.4.006

- (a) $3\sqrt{3}$ — Extract square factors.
 (b) $\sqrt{7/7}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
 (c) 3 — Difference of squares.

T.III.4.007

- (a) 7.9×10^{-3} — $1 \leq A < 10$.
 (b) 340000 — Shift decimal.
 (c) 1.264×10^0 — Multiply A; add exponents; renormalise.

T.III.4.008

- (a) $4\sqrt{2}$ — Extract square factors.
 (b) $\sqrt{3/3}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
 (c) -1 — Difference of squares.

T.III.4.009

- (a) $3\sqrt{3}$ — Extract square factors.
 (b) $\sqrt{7/7}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
 (c) 3 — Difference of squares.

T.III.4.010

- (a) 2.7×10^5 — $1 \leq A < 10$.
 (b) 30000 — Shift decimal.
 (c) 6.75×10^7 — Multiply A; add exponents; renormalise.

T.III.4.011

- (a) $2\sqrt{2}$ — Extract square factors.
 (b) $\sqrt{7/7}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
 (c) 3 — Difference of squares.

T.III.5.001

- (a) $2x + 8$ — Distribute.
 (b) $x^2 + 10x + 25$ — FOIL.
 (c) $(x+5)(x+5)$ — Factor pair.

T.III.5.002

- (a) $4x + 20$ — Distribute.
 (b) $x^2 + 6x + 9$ — FOIL.
 (c) $(x+3)(x+3)$ — Factor pair.

T.III.5.003

- (a) $4x + 16$ — Distribute.
 (b) $x^2 + 7x + 12$ — FOIL.
 (c) $(x+4)(x+3)$ — Factor pair.

T.III.5.004

- (a) $2\sqrt{2}$ — Extract square factors.
 (b) $\sqrt{3/3}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
 (c) -1 — Difference of squares.

T.III.5.005

- (a) $2x + 6$ — Distribute.
 (b) $x^2 + 6x + 8$ — FOIL.
 (c) $(x+2)(x+4)$ — Factor pair.

T.III.5.006

- (a) $3x + 6$ — Distribute.
 (b) $x^2 + 5x + 6$ — FOIL.
 (c) $(x+2)(x+3)$ — Factor pair.

T.III.5.007

- (a) $3x + 6$ — Distribute.
 (b) $x^2 + 4x + 3$ — FOIL.
 (c) $(x+1)(x+3)$ — Factor pair.

T.III.5.008

- (a) $5x + 5$ — Distribute.
 (b) $x^2 + 9x + 18$ — FOIL.
 (c) $(x+3)(x+6)$ — Factor pair.

T.III.5.009

- (a) $5x + 5$ — *Distribute.*
- (b) $x^2 + 3x + 2$ — *FOIL.*
- (c) $(x+1)(x+2)$ — *Factor pair.*

T.III.5.010

- (a) $5x + 20$ — *Distribute.*
- (b) $x^2 + 3x + 2$ — *FOIL.*
- (c) $(x+1)(x+2)$ — *Factor pair.*

T.III.5.011

- (a) $6x + 6$ — *Distribute.*
- (b) $x^2 + 6x + 8$ — *FOIL.*
- (c) $(x+4)(x+2)$ — *Factor pair.*

T.III.6.001

- (a) $x = 5$ — *Isolate x.*
- (b) $x = 6$ — *Expand/divide.*
- (c) 6, 7, 8, 9 — *Integer solutions of inequality.*

T.III.6.002

- (a) $x = 6$ — *Isolate x.*
- (b) $x = 7$ — *Expand/divide.*
- (c) 7, 8, 9, 10 — *Integer solutions of inequality.*

T.III.6.003

- (a) $x = 3$ — *Isolate x.*
- (b) $x = 4$ — *Expand/divide.*
- (c) 4, 5, 6, 7 — *Integer solutions of inequality.*

T.III.6.004

- (a) $x = 9$ — *Isolate x.*
- (b) $x = 10$ — *Expand/divide.*
- (c) 10, 11, 12, 13 — *Integer solutions of inequality.*

T.III.6.005

- (a) $x = 5$ — *Isolate x.*
- (b) $x = 6$ — *Expand/divide.*
- (c) 6, 7, 8, 9 — *Integer solutions of inequality.*

T.III.6.006

- (a) $x = 5$ — *Isolate x.*
- (b) $x = 6$ — *Expand/divide.*
- (c) 6, 7, 8, 9 — *Integer solutions of inequality.*

T.III.6.007

- (a) $x = 9$ — *Isolate x.*
- (b) $x = 10$ — *Expand/divide.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.III.6.008

- (a) $x = 5$ — *Isolate x.*
- (b) $x = 6$ — *Expand/divide.*
- (c) 6, 7, 8, 9 — *Integer solutions of inequality.*

T.III.6.009

- (a) $x = 9$ — *Isolate x.*
- (b) $x = 10$ — *Expand/divide.*
- (c) 10, 11, 12, 13 — *Integer solutions of inequality.*

T.III.6.010

- (a) $x = 1$ — *Isolate x.*
- (b) $x = 2$ — *Expand/divide.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.III.6.011

- (a) $x = 4$ — *Isolate x.*
- (b) $x = 5$ — *Expand/divide.*
- (c) 5, 6, 7, 8 — *Integer solutions of inequality.*

T.III.7.001

- (a) $2x + 10$ — *Distribute.*
- (b) $x^2 + 7x + 12$ — *FOIL.*

(c) $(x+3)(x+4)$ — Factor pair.

T.III.7.002

(a) $2x + 2$ — Distribute.

(b) $x^2 + 11x + 30$ — FOIL.

(c) $(x+5)(x+6)$ — Factor pair.

T.III.7.003

(a) $6x + 36$ — Distribute.

(b) $x^2 + 3x + 2$ — FOIL.

(c) $(x+2)(x+1)$ — Factor pair.

T.III.7.004

(a) $5x + 10$ — Distribute.

(b) $x^2 + 9x + 20$ — FOIL.

(c) $(x+4)(x+5)$ — Factor pair.

T.III.7.005

(a) $4x + 4$ — Distribute.

(b) $x^2 + 5x + 6$ — FOIL.

(c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.III.7.006

(a) $5x + 30$ — Distribute.

(b) $x^2 + 10x + 25$ — FOIL.

(c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.III.7.007

(a) $4x + 20$ — Distribute.

(b) $x^2 + 5x + 4$ — FOIL.

(c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.III.7.008

(a) $4x + 20$ — Distribute.

(b) $x^2 + 9x + 18$ — FOIL.

(c) $(x+3)(x+6)$ — Factor pair.

T.III.7.009

(a) $2x + 10$ — Distribute.

(b) $x^2 + 5x + 4$ — FOIL.

(c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.III.7.010

(a) $5x + 15$ — Distribute.

(b) $x^2 + 9x + 20$ — FOIL.

(c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.III.7.011

(a) $5x + 25$ — Distribute.

(b) $x^2 + 3x + 2$ — FOIL.

(c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.III.8.001

(a) Top view (from above) — Plan = bird's-eye.

(b) 6 — Six faces.

(c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.III.8.002

(a) 6 — One face per side.

(b) Row of 6 cannot (exceeds max chain length 4) — Cube-net rules.

(c) $292 - 2(lw+wh+hl)$.

T.III.8.003

(a) 255 — Pad to 3 digits; clockwise from north.

(b) 075 — $\pm 180^\circ$.

(c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.III.8.004

(a) Top view (from above) — Plan = bird's-eye.

(b) 6 — Six faces.

(c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.III.8.005

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.III.8.006

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.III.8.007

- (a) 10 cm — $\sqrt{a^2+b^2}$.
- (b) 8 m — $\sqrt{c^2-a^2}$.
- (c) 8 m — *Pythagoras.*

T.III.8.008

- (a) 10 cm — $\sqrt{a^2+b^2}$.
- (b) 8 m — $\sqrt{c^2-a^2}$.
- (c) 8 m — *Pythagoras.*

T.III.8.009

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.III.8.010

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.III.8.011

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.III.8.012

- (a) 9 — SF^2 .
- (b) 18 cm — $\times SF$.
- (c) 27 — SF^3 .

T.III.8.013

- (a) SSS, SAS, ASA (and RHS) — *Standard conditions.*
- (b) No — ambiguous case possible — *SSA not a congruence condition.*
- (c) Corresponding parts of congruent triangles are congruent — *CPCTC.*

T.III.8.014

- (a) 25 — SF^2 .
- (b) 30 cm — $\times SF$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.III.8.015

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.III.8.016

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.III.8.017

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.III.8.018

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.III.8.019

- (a) $1/2$ — *Special triangles.*
- (b) $\sqrt{2}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.III.8.020

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.III.8.021

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.III.8.022

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.III.8.023

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.III.8.024

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.III.8.025

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.III.8.026

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.III.8.027

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.III.8.028

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.III.8.029

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.III.9.001

- (a) $2/3$ — $8/12$.
- (b) $14/33$ — *Product along tree.*
- (c) $2/9$ — *Independent draws.*

T.III.9.002

- (a) $3/5$ — $3/5$.
- (b) $3/10$ — *Product along tree.*
- (c) $6/25$ — *Independent draws.*

T.III.9.003

- (a) 11.5 — *Sum $69 \div 6$.*
- (b) 14 — *max-min.*
- (c) 19 — $5 \times \text{mean} - \text{sum of four}$.

T.III.9.004

- (a) 3.0 — *$FD = \text{frequency} \div \text{class width}$.*
- (b) 15 — *$\text{frequency} = FD \times \text{width}$.*

(c) So area is proportional to frequency — *Area encoding*.

T.III.9.005

- (a) 8.33 — *Sum* $50 \div 6$.
- (b) 9 — *max–min*.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.III.9.006

- (a) 7.33 — *Sum* $44 \div 6$.
- (b) 11 — *max–min*.
- (c) 39 — $5 \times \text{mean}$ — *sum of four*.

T.III.9.007

- (a) 11.5 — *Sum* $69 \div 6$.
- (b) 12 — *max–min*.
- (c) 23 — $5 \times \text{mean}$ — *sum of four*.

T.III.9.008

- (a) 9.17 — *Sum* $55 \div 6$.
- (b) 9 — *max–min*.
- (c) 15 — $5 \times \text{mean}$ — *sum of four*.

T.III.9.009

- (a) 9.17 — *Sum* $55 \div 6$.
- (b) 10 — *max–min*.
- (c) 15 — $5 \times \text{mean}$ — *sum of four*.

T.III.9.010

- (a) 9.67 — *Sum* $58 \div 6$.
- (b) 8 — *max–min*.
- (c) 21 — $5 \times \text{mean}$ — *sum of four*.

T.III.9.011

- (a) 12.17 — *Sum* $73 \div 6$.
- (b) 14 — *max–min*.
- (c) 33 — $5 \times \text{mean}$ — *sum of four*.

T.III.9.012

- (a) 7.5 — *Sum* $45 \div 6$.
- (b) 11 — *max–min*.
- (c) 50 — $5 \times \text{mean}$ — *sum of four*.

T.III.9.013

- (a) 11.83 — *Sum* $71 \div 6$.
- (b) 11 — *max–min*.
- (c) 15 — $5 \times \text{mean}$ — *sum of four*.

T.III.9.014

- (a) 10.33 — *Sum* $62 \div 6$.
- (b) 11 — *max–min*.
- (c) 19 — $5 \times \text{mean}$ — *sum of four*.

T.III.9.015

- (a) 7.67 — *Sum* $46 \div 6$.
- (b) 11 — *max–min*.
- (c) 31 — $5 \times \text{mean}$ — *sum of four*.

T.III.9.016

- (a) 12.17 — *Sum* $73 \div 6$.
- (b) 15 — *max–min*.
- (c) 27 — $5 \times \text{mean}$ — *sum of four*.

T.III.9.017

- (a) 10.5 — *Sum* $63 \div 6$.
- (b) 13 — *max–min*.
- (c) 8 — $5 \times \text{mean}$ — *sum of four*.

T.III.9.018

- (a) 9.83 — *Sum* $59 \div 6$.
- (b) 14 — *max–min*.
- (c) 17 — $5 \times \text{mean}$ — *sum of four*.

T.III.9.019

- (a) The governing definition/theorem for Probability scale 0 to 1 language — *Name it precisely.*
 (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
 (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.III.9.020

- (a) 7.33 — *Sum $44 \div 6$.*
 (b) 7 — *max–min.*
 (c) 42 — *$5 \times \text{mean}$ – sum of four.*

T.III.9.021

- (a) 13.83 — *Sum $83 \div 6$.*
 (b) 9 — *max–min.*
 (c) 22 — *$5 \times \text{mean}$ – sum of four.*

T.III.9.022

- (a) The governing definition/theorem for AP Stats: comparing dual boxplots FRQ — *Name it precisely.*
 (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
 (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.III.9.023

- (a) 11.67 — *Sum $70 \div 6$.*
 (b) 15 — *max–min.*
 (c) 10 — *$5 \times \text{mean}$ – sum of four.*

T.III.9.024

- (a) 8.67 — *Sum $52 \div 6$.*
 (b) 14 — *max–min.*
 (c) 27 — *$5 \times \text{mean}$ – sum of four.*

T.III.9.025

- (a) 8.33 — *Sum $50 \div 6$.*
 (b) 9 — *max–min.*
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.III.9.026

- (a) 7.5 — *Sum $45 \div 6$.*
 (b) 12 — *max–min.*
 (c) 50 — *$5 \times \text{mean}$ – sum of four.*

T.III.9.027

- (a) 11.83 — *Sum $71 \div 6$.*
 (b) 14 — *max–min.*
 (c) 3 — *$5 \times \text{mean}$ – sum of four.*

T.III.9.028

- (a) 2.4 — *$FD = \text{frequency} \div \text{class width}$.*
 (b) 24 — *$\text{frequency} = FD \times \text{width}$.*
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.III.9.029

- (a) 10.33 — *Sum $62 \div 6$.*
 (b) 12 — *max–min.*
 (c) 14 — *$5 \times \text{mean}$ – sum of four.*

T.III.9.030

- (a) 10.83 — *Sum $65 \div 6$.*
 (b) 15 — *max–min.*
 (c) 25 — *$5 \times \text{mean}$ – sum of four.*

T.III.9.031

- (a) 13.33 — *Sum $80 \div 6$.*
 (b) 10 — *max–min.*
 (c) 14 — *$5 \times \text{mean}$ – sum of four.*

T.III.9.032

- (a) $7/9$ — *$7/9$.*
 (b) $7/12$ — *Product along tree.*
 (c) $14/81$ — *Independent draws.*

T.III.9.033

- (a) $3/5$ — $3/5$.
- (b) $3/10$ — *Product along tree.*
- (c) $6/25$ — *Independent draws.*

T.III.9.034

- (a) $4/5$ — $8/10$.
- (b) $28/45$ — *Product along tree.*
- (c) $4/25$ — *Independent draws.*

T.III.9.035

- (a) 1.1 — $FD = \text{frequency} \div \text{class width}$.
- (b) 22 — $\text{frequency} = FD \times \text{width}$.
- (c) So area is proportional to frequency — *Area encoding.*

T.III.1.021

- (a) $55 \leq L < 65$ — $\pm$ *half unit.*
- (b) 28.40 cm — *Use UBs in $P=2(L+W)$.*
- (c) $UB(\text{distance}) \div LB(\text{time})$ — *Max quotient: max numerator, min denominator.*

T.III.3.022

- (a) $\pounds 230$ — $\times 1.15$.
- (b) $\pounds 199.13$ — $\div 1.15$.
- (c) 32.25% — *Multiply multipliers; convert to %.*